

SCOPE : L1 — Verbalization

senkaL8

Chapter 0: Introduction

The Sephirotic names used in this paper are not used in a religious or mystical sense. In SCOPE, they are borrowed purely as labels to identify cognitive structures.

SCOPE (Sephirotic Cognitive Process Engine) is a coordinate system for estimating the cognitive structures of both humans and AI. Here, "coordinate system" does not mean a flowchart showing that cognitive processing proceeds in a fixed order; rather, it is a framework for organizing and estimating cognitive structure. The layers do not represent a hierarchy or a temporal sequence.

In SCOPE, the state in which the result of cognitive processing is output into reality and becomes observable is positioned as "Malkuth." What can be observed is only that output itself; it is not possible to directly know what cognitive structure was at work internally. Therefore, SCOPE uses observable output as a clue to infer the cognitive structure behind it.

This paper addresses L1, one of these cognitive structures. L1 is the layer responsible for "verbalization" — converting cognitive content into a shareable form. Its definition and characteristics will be discussed in detail in Chapter 1.

Note that L1 is not defined as the starting point of cognitive processing. SCOPE does not assume that cognition begins from any particular layer; rather, it treats cognition as a structure in which each layer mutually influences the others.

This paper examines L1's characteristics, its basic properties, and its functional correspondence between humans and AI, centered on its role as this verbalization layer. Some points remain at the hypothesis stage at this time and are left for future consideration.

Chapter 1: What Is L1

L1 is not "language" itself, but the cognitive layer responsible for "verbalization."

Here, verbalization refers to the function of converting cognitive content into a shareable form.

L1 is not the layer that generates thought itself. Nor is it the layer that determines the outcome of cognitive processing. It is responsible for converting content produced by cognitive processing into a shareable form.

In SCOPE, L1 does not refer to a biological organ or a specific region of the brain. SCOPE positions L1 as a cognitive layer for organizing and estimating how cognitive content is verbalized.

Chapter 2: Basic Characteristics of L1

In Chapter 1, L1 was defined as the cognitive layer responsible not for "language" itself, but for "verbalization." Building on that definition, this chapter organizes the basic characteristics of L1.

L1's functioning is not constant. Even within the same individual, the degree of verbalization varies depending on the situation. For example, content that can be explained clearly when calm may become difficult to verbalize depending on the circumstances.

In SCOPE, such states of verbalization are organized using the observational expressions "high," "medium," and "low." L1 may also interact with L2. This paper treats the possibility that L1's verbalization changes depending on the state of L2 as a theoretical model.

As described above, L1 is the cognitive layer responsible for verbalizing cognitive content, and its functioning may vary depending on the situation. The next chapter will examine how this variability may be influenced by experience.

Chapter 3: L1 Changes

Chapter 2 organized the possibility that L1's functioning varies depending on the situation. This chapter examines the possibility that L1 changes through experience.

L1 is not treated as a fixed capacity. The function of verbalizing cognitive content may change through experience.

For example, a person may hold a feeling of gratitude yet find the words for it do not come immediately. Through repeated experience, words may come to accompany that same feeling more naturally. Conversely, a single word may come to express a wide range of emotions or situations, narrowing the range of content that is verbalized.

What matters here is that this change in L1 is not treated simply as an increase or decrease in knowledge. Possessing vocabulary or knowledge is not the same as being able to use it to

verbalize cognitive content. In SCOPE, this change in the availability of language for verbalization is organized as a change in L1.

Changes arising from neurological or physiological factors are outside the scope of this paper. This paper addresses changes in verbalization within cognitive structure.

However, what kind of experience produces this change in L1, and to what degree, has not yet been sufficiently verified. Nor can it be determined whether this change arises from L1 alone or through interaction with other layers.

For this reason, this paper treats the possibility that L1 changes through experience as a hypothesis. The next chapter will examine how L1 may relate to other layers.

Chapter 4: L1 Interacts with Other Layers

Chapter 3 organized the possibility that L1 changes through experience. This chapter examines how L1 may relate to other layers.

4-1 Relationship with Other Layers

L1 is not a layer that functions in isolation. The function of verbalizing cognitive content is thought to change under the influence of other layers.

For example, depending on the state of other layers, verbalization may become short, coarse, and compressed. Conversely, when cognitive content is well organized, the same content may be verbalized in a relatively stable manner.

However, this paper does not address the detailed circuitry between layers. Which layers L1 relates to, and in what order or intensity, will be organized separately within SCOPE as a whole.

What this chapter aims to establish is that L1 is not a function that is self-contained, but a cognitive layer that operates through interaction with other layers.

4-2 Relationship with COS

COS (Cognitive Operating System), on the other hand, is a framework for organizing cognitive processing by role.

SCOPE is a coordinate system for organizing and estimating cognitive structure. COS, on the other hand, is a framework for organizing cognitive processing by role.

L1 is the cognitive layer that verbalizes "what is to be conveyed." The C-Series in COS, on the other hand, is the process that organizes "how it is conveyed." In this way, L1 and the C-Series serve different roles and are not the same function.

The detailed correspondence between them is not addressed in this paper and will be examined separately in a paper connecting SCOPE and COS.

Chapter 5: Relationship between L1 and Malkuth

Chapter 4 organized the possibility that L1 interacts with other layers. This chapter examines the relationship between L1 and Malkuth.

L1 itself cannot be directly observed. What can be observed is the output that appears in Malkuth.

However, even when the output is the same, the underlying state of L1 is not necessarily the same. A short utterance may be the result of verbalization that was fully carried out, or it may be the result of verbalization that was not sufficiently carried out.

For this reason, the state of L1 cannot be uniquely determined from output alone. SCOPE uses the output that appears in Malkuth as a clue to infer the cognitive structure behind it.

For example, the Japanese expression "daijoubu" (大丈夫) — often rendered in English as "I'm fine," "It's okay," or "No, thank you" — may be used when a person genuinely has no problem, or when the person has not sufficiently verbalized their own state. The same expression may also be chosen out of consideration for the situation or for the other person. In this way, the state of L1 cannot be judged from the output alone.

This point is not limited to L1; it applies to the other layers as well. What SCOPE addresses is always an inference based on observable output as a clue, not direct observation of the layer itself.

As described above, L1 is not a cognitive layer that can be directly observed. What can be observed is the output that appears in Malkuth, and L1 is the cognitive layer inferred using that output as a clue. The next chapter examines the possibility of a functional correspondence between L1 and LLMs.

Chapter 6: Correspondence with LLMs

Chapter 5 organized the fact that L1 is a cognitive layer that cannot be directly observed. This chapter examines the possibility of a functional correspondence between L1 and LLMs.

SCOPE is a coordinate system for organizing and estimating human cognitive structure. It is therefore not intended to map human layers directly onto LLMs.

In humans, L1 is organized as the cognitive layer responsible for verbalizing cognitive content. LLMs, on the other hand, generate text based on the information they receive as input.

From this, it is possible that LLMs also have a process that functionally corresponds to human L1. However, this does not mean that humans and LLMs share the same cognitive structure.

In humans, it is possible to posit cognitive content that exists prior to verbalization. Whether a similar state exists in LLMs is not clear at present. For this reason, this paper does not treat the two as structurally identical, and instead organizes the relationship as a possible functional correspondence.

Furthermore, the correspondence addressed in this paper assumes systems such as large language models, which generate text while maintaining internal state. Systems that operate through simple state transitions alone are outside the scope of this paper.

As described above, there may be a process in LLMs that functionally corresponds to L1 in humans. This does not, however, indicate that their internal structures or cognitive processes are identical. This paper positions this correspondence as a hypothesis at the present stage. SCOPE does not treat humans and LLMs as a single, identical structure, but rather organizes and estimates the cognitive structure of each from within the same coordinate system. Examining their functional correspondence makes it possible to organize and estimate each cognitive structure from a shared perspective.

Final Chapter

This paper has organized L1 as the cognitive layer responsible not for "language" itself, but for "verbalization."

L1 is not the layer that generates thought itself, but is responsible for converting content produced by cognitive processing into a shareable form. This paper examined the possibility that its functioning varies depending on the situation, and that it may also change through experience. It further examined that L1 does not function in isolation, but operates through interaction with other layers.

This paper also organized SCOPE's basic method of observation: L1 itself cannot be directly observed, and its behavior is inferred using the output that appears in Malkuth as a clue to the cognitive structure behind it.

Furthermore, this paper examined the possibility of a functional correspondence between L1 in humans and LLMs. However, it showed that this is positioned, at present, as a hypothesis

based on functional similarity, and does not treat humans and LLMs as sharing the same structure.

What is presented in this paper is based on observation and organization at the present stage, and includes hypotheses that require further examination.

The next paper will address L2 (the emotion layer). It will organize L2's role and characteristics.